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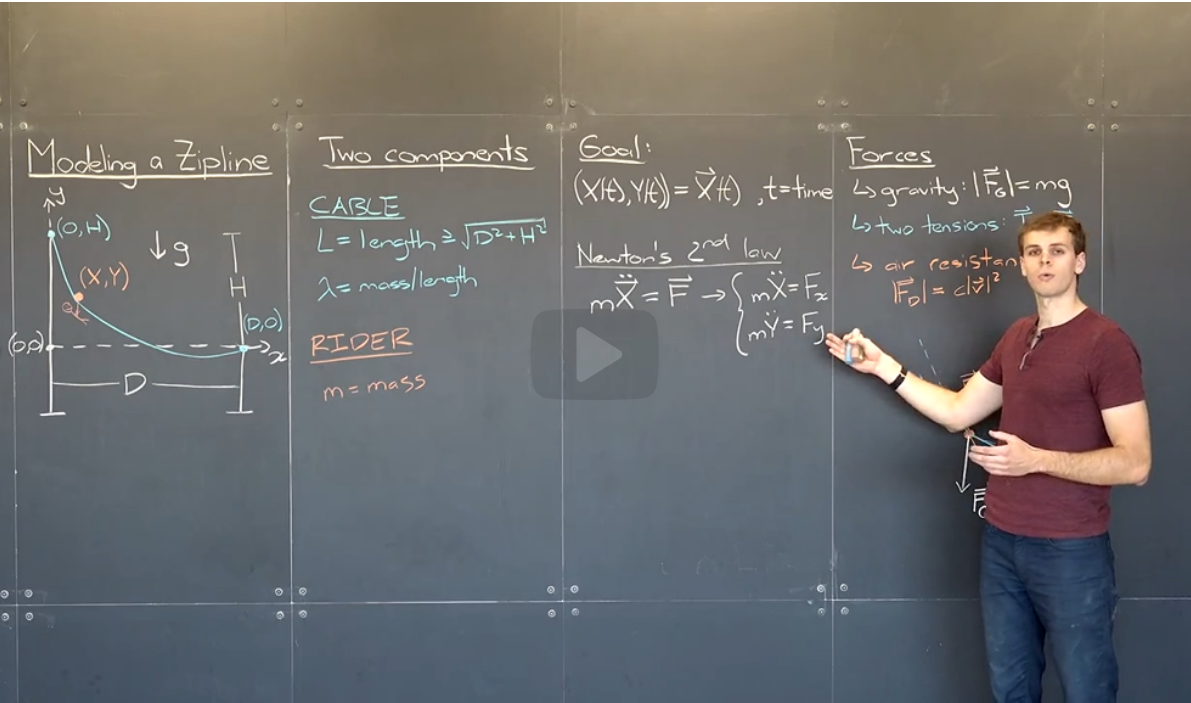


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Introducing a new problem



the form of what is called a catenary. We'll then see that when a rider is traveling along the cable, we can approximate the shape of the cable by two catenaries on either side of the rider. Finally, we'll put together everything we've learned to solve these differential equations and obtain the function [vector] $\mathbf{X}(t)$.

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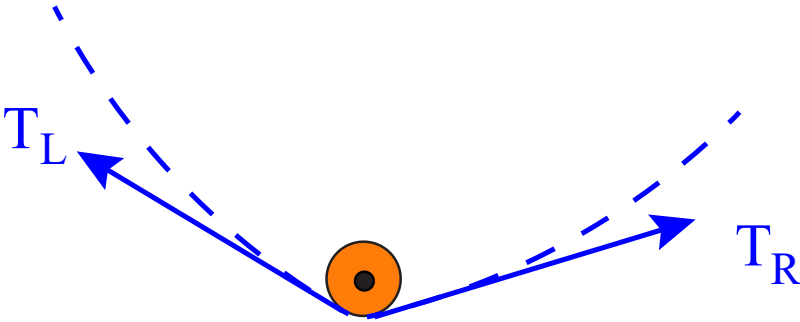
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A simplification using torque balance

Note that the modeling of this system is much more complicated than any other system we have modeled thus far. One reason for this is the pulley which is what moves the rider along the cable. Because the mass of the pulley is negligible, the torques on the pulley must balance. What are the forces which may contribute to the torque?



There are two tension forces $\mathbf{T}_L$ and $\mathbf{T}_R$ tangent to the outside edge of the pulley pointing along the cable to the left and right of the pulley.

If we take the origin of the pulley to be the center of the pulley, then the forces exerted on the pulley by the rider are applied to the center and do not contribute to the torque. In this model, we will ignore friction in the pulley. Thus the only two forces contributing to the torque are the two tension forces. The tension forces have different directions at the same radial distance from the center of the pulley. Thus torque balance tells us that the magnitude of these two tension forces must be equal. We write the magnitude as T , setting

$$T = |\mathbf{T}_L| = |\mathbf{T}_R|.$$

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